

Teaching Research Integrity through Verification of AI-Generated References: An Activity for Upper-Level Chemistry Courses

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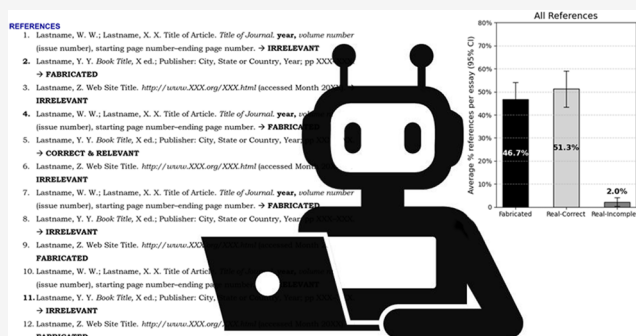
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ABSTRACT: Integrating generative AI (GenAI) into chemistry coursework is transforming how students approach scientific writing and evaluate information. In this activity, implemented in upper-level biochemistry and bioinorganic chemistry courses, 69 undergraduate and graduate students used chatbots like ChatGPT, U-M GPT, and Claude to create bibliographies for essays on chemistry research topics. Students then critically engaged with the chemical literature, systematically validating each AI-generated citation using scholarly tools. Our analysis of 456 GenAI-generated references revealed that on average a student essay contained 51.3% real and correctly cited references. These authentic citations were most frequently associated with multidisciplinary high-impact journals, particularly those in the *Nature* portfolio, and were linked to articles with high Altmetric Attention Scores. Through this process, students directly encountered the challenges and pitfalls of relying on AI-generated sources. Especially concerning was the observation that fabricated references were most frequently associated with disciplinary chemistry journals. Survey analysis and in-class discussions following the GenAI-assisted writing activity revealed that students developed a greater awareness of research integrity and gained practical skills for critically assessing scholarly sources. For example, when evaluating references from chemistry journals, students often identified inconsistencies and relied on their domain-specific knowledge to correct citations or locate original articles. Of the original 456 AI-generated citations, only 58% remained in students' final essays, underscoring ongoing difficulties with both citation accuracy and relevance and highlighting the critical importance of source evaluation when working with AI. Overall, this activity underscores the importance of fostering both AI literacy and research integrity in chemistry education and offers actionable recommendations to help students use AI responsibly in academic writing.

KEYWORDS: General Public, Upper-Division Undergraduate, Graduate Education/Research, Communication/Writing, Computer-Based Learning, Biochemistry, Nucleic Acids, Interdisciplinary/Multidisciplinary



INTRODUCTION

The rapid adoption of generative artificial intelligence (GenAI) technologies, particularly those based on large language models (LLMs), such as ChatGPT (Generative Pretrained Transformer by OpenAI),¹ has begun to reshape the academic landscape. Since its public release at the end of 2022, ChatGPT and similar GenAI-tools have been widely used by academics for research, teaching, and administrative tasks.² GenAI tools have become increasingly popular for academic writing and editing, not only among students but also among senior academics, offering quick and accessible support that saves time and helps reduce academic stress.^{3–5} In the context of chemical education, GenAI-assisted writing assignments are becoming a common practice.^{6–11} Additionally, chemistry educators have been exploring the use of GenAI technologies to create instructional materials, support student learning, facilitate group activities, evaluate student works, and personalize learning experiences.^{12–21} Recognizing both the inevitability and the benefits of

AI, the U.S. Department of Education has encouraged educational institutions to use artificial intelligence to enhance students' learning outcomes.²² The Department of Education emphasizes that AI has the potential to foster personalized learning, strengthen critical thinking, and prepare students for complex problem-solving of logical tasks.

However, the adoption of GenAI technology remains controversial in academia, mainly due to valid concerns that it may compromise research integrity, stifle creativity, hinder the development of critical thinking skills, and ultimately lower scholarly standards.^{19,23–30} An exemplary issue is the proper

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attribution of credit, a fundamental scholarly practice. Although AI tools can generate well-formatted bibliographies, accumulating evidence demonstrates that GenAI chatbots can fabricate citations or introduce substantial errors in bibliographic details.^{7,31–38} This issue was highlighted in a widely publicized legal case in which a U.S. federal judge was presented with nonexistent, AI-generated court citations that an attorney had unknowingly submitted, believing them to be legitimate.³⁹ As a result, the court considered imposing sanctions for the improper use of generative AI. Given the ability of GenAI systems to produce convincing, yet fictitious, citations, it is crucial for students to not only harness the benefits of these tools but also to critically evaluate the reliability of the information they produce.

While discussions of research integrity in science frequently focus on issues surrounding irreproducibility of results and scientific misconduct,^{40–42} ethical considerations in regard to reference practices are often overlooked. Yet, the integrity of a reference list is fundamental in academic work because it shows where your ideas come from and helps prevent misattribution or plagiarism. Providing effective instruction in citation practices is therefore essential, and not only for ensuring that students cite scholarly literature responsibly, but also for cultivating the ethical standards expected within the research community. Additionally, citations affect how scholarly articles are indexed and how research impact is evaluated.⁴³ Inaccurate referencing can disrupt the logical progression of ideas, skew the results of systematic reviews, and potentially lead to incorrect conclusions. Instilling strong habits in building accurate bibliographies, along with the skills needed to locate, assess, and accurately attribute sources, not only strengthens information literacy skills but also fosters a culture of research integrity.

Prior studies in chemistry education have demonstrated that incorporating citation analysis in teaching assignments helps students gain a deeper understanding of the research topic, appreciate its current state-of-the-art in regard to the prior knowledge, develop essential skills in information-gathering and validation, and understand the motivations behind researchers' citation choices.^{44–51} Given the importance of these skills, it is no surprise that the American Chemical Society's Guidelines and Evaluation Procedures for Bachelor's Degree Programs⁵² and the Association of College & Research Libraries (ACRL) Information Literacy Standards for Science and Engineering/Technology⁵³ recommend that educators implement assignments that focus on developing students' competencies in assessing the quality of scientific information and proficiency in searching, reading, analyzing, interpreting, and citing information sources. In an era where artificial intelligence increasingly shapes how students interact with scholarly materials, teaching students how to critically examine and interpret the GenAI-generated information sources is essential for cultivating strong habits of academic rigor and ethical integrity.

Recognizing these challenges and opportunities, we developed and implemented a writing-based assignment in advanced undergraduate and graduate chemistry courses to emphasize research integrity through the process of evaluating AI-generated references. The UNESCO AI Competency Frameworks^{54,55} recommend that student education in AI technologies should combine technical proficiency with ethical awareness and critical reflection. Accordingly, our classroom assignment extends beyond basic source-finding to include validation and critical assessment of academic sources. By making the evaluation of AI-fabricated references an explicit, structured,

and reflective component of students' academic experience, this activity transforms research integrity from an abstract concept into a tangible, everyday practice. Here we present the activity structure, reference analysis, and perceived student outcomes, alongside practical recommendations for integrating this type of activity into chemistry curricula.

METHODS

Instructional Design

This study is based on writing assignments submitted by students enrolled in three different upper-level chemistry courses offered in Fall 2023 and 2024; namely, two semesters of Nucleic Acid Biochemistry and one semester of Bioinorganic Chemistry. The cohort comprised 80% ($N = 55$) undergraduate students, of which 89% ($N = 49$) majored in biological sciences or biochemistry. The majority of graduate students specialized in chemical biology (64%, $N = 9$), others indicated inorganic or organic chemistry as areas of expertise.

Sixty-nine individual essays on distinct topics related to bio- or bioinorganic chemistry (see Figure S1) were produced as part of the writing assignment described earlier.⁷ Briefly, the assignment included two main components: (a) a GenAI-assisted draft of 500 words that contained in-text citations and a bibliography and (b) a final essay of 1000 words, for which students reviewed the GenAI-assisted draft for accuracy, completeness, and flow and expanded the content by adding their own perspectives and ensuring accuracy and relevance of citations.

The assignment was supported by library instructions that covered the following topics:⁵⁶

1. Basic operational principles of LLMs with an emphasis on ChatGPT and U-M GPT; the latter is a generative AI chatbot specifically developed for the University of Michigan community.⁵⁷
2. Common pitfalls associated with GenAI, such as copyright violations, environmental impacts, hallucination issues, and various biases.
3. Prompt literacy and corresponding training in developing effective prompts to maximize the usefulness and accuracy of GenAI tools. An example of student's designed prompts for creating a bibliography: "Include an in-text citation for each topic discussed and a bibliography at the end referencing each of the in-text citations."
4. Reference validation strategies using scholarly databases (e.g., PubMed).⁵⁸

Data Analysis

This study analyzes the difference between the initial bibliographies in GenAI-assisted drafts and the final bibliographies submitted by students with their final essays. A total of 456 references were collected and analyzed (an average of 6.6 references per student).

Each citation was inspected for authenticity using a combination of Google Scholar, PubMed, and Dimensions Plus databases. Dimensions Plus is a comprehensive scholarly database that additionally provides Altmetric Attention Scores, which measure the social media visibility and impact of individual publications.⁵⁹ To verify each reference, we initially searched using the AI-generated DOI or link. If a DOI or link was not available, the search was conducted based on title, journal name, and authors' names.

For each reference, six components were tracked: (i) title, (ii) authors, (iii) journal name, (iv) year of publication, (v) volume and/or issue, and (vi) page numbers, marking them as correct or incorrect (Figure 1). For a title to be considered correct, the title needed to be nearly identical (allowing variations in stop words, punctuation, and 1–2 content words), and for the author to be correct at least one author in the list had to be accurately cited. Additional variables recorded included the AI tool used, essay topic, article Altmetric score, number of times the article was cited, Wikipedia mentions, DOI, and the prompt used to generate the bibliography. All data were manually extracted from student essays and compiled into a newly created. csv data set. This enabled subsequent analysis and visualization in Python using

RNA Therapeutics for Neurodegenerative Diseases: Promising Approaches to Target RNA Molecules

References (Chat GPT, GPT 3.5)

Baden, L. R., El Sahly, H. M., Essink, B., Kotloff, K., Frey, S., Novak, R., ... & Zaks, T. (2020). Efficacy and Safety of the mRNA-1273 SARS-CoV-2 Vaccine. *New England Journal of Medicine*, 384(5), 403-416. ✖

Hodi, F. S., O'Day, S. J., McDermott, D. F., Weber, R. W., Sosman, J. A., Haanen, J. B., & Boasberg, P. D. (2016). Improved survival with ipilimumab in patients with metastatic melanoma. *New England Journal of Medicine*, 363(8), 711-723. ✖

Lan, X., Wang, W., Li, Q., Wang, J., & Chen, M. (2019). RNA-based immunotherapy for metastatic clear cell renal cell carcinoma. *Journal of Immunology Research*. ⚠

Li, J., Sun, Y., Jia, T., Zhang, R., Zhang, K., Wang, L., & Wei, W. (2021). Messenger RNA vaccine strategies for cancer immunotherapy. *npj Vaccines*, 6(1), 1-10. ⚠

Sahin, U., Derhovanessian, E., Miller, M., Kloeke, B. P., Simon, P., Lower, M., ... & Tadmor, A. D. (2017). Personalized RNA melanoma vaccines mobilize poly-specific therapeutic immunity against cancer. *Nature*, 547(7662), 222-226. ✖

Weissman, D., Karikó, K., & de Fougères, A. (2020). mRNA: Fulfilling the Promise of Gene Therapy. *Molecular Therapy*, 27(4), 705-707. ✔

Wittrup, A., & Lieberman, J. (2015). Knocking down disease: a progress report on siRNA therapeutics. *Nature Reviews Genetics*, 16(9), 543-552. ✔

Figure 1. Example of GenAI-generated references, marked according to their accuracy and relevance. For accuracy, each reference is evaluated across six citation components: author, year, title, journal, volume/issue, and page numbers. These components are highlighted in yellow if they are real and correctly cited or in gray if they are fabricated by GenAI. At the end of each reference, a symbol indicates its status: a green checkmark for Relevant and Verified (included in the final essay draft); a red ✖ for Irrelevant (omitted from the final draft); or an orange triangle for Included but Not Verified (included in the final essay despite questionable authenticity).

libraries such as seaborn, pandas, and numpy within Jupyter Notebook. Descriptive analyses and Student's *t* tests were conducted using the Statistical Package for the Social Sciences (SPSS), version 30.

Data Categorization

The extracted references were classified according to their accuracy and relevance (Figure 1).

1. Reference accuracy:

- Real and Correctly Cited (RCC) references ($N = 245$): all six reference components were correct.

- Real but Incomplete (RI) references ($N = 7$): up to two of the six reference components were missing, while the remaining components were correct. Typical missing components included pages, volume, or issue.
- Fabricated Citations (FC) ($N = 204$): at least one component of the reference was incorrect. Of these, 48% ($N = 98$) were completely unidentifiable, meaning they could not be matched to any existing publication.

2. Reference relevance:

- Relevant references ($N = 265$): the original GenAI-generated reference that a student-author decided to keep in their final essay (including a fabricated reference).
- Irrelevant reference ($N = 191$): the original GenAI-generated reference that a student-author decided to remove from their final essay.

Assessment of Student Experiences

Student perceptions regarding the evaluation of AI-generated references were collected via an anonymous survey. The University of Michigan Health Sciences and Behavioral Sciences Institutional Review Board classified this research as "exempt" (HUM00245013). The survey methodology, including questions, coding procedure, and statistical analysis, is detailed in a separate publication,⁶⁰ while specific information relevant to this study is provided in Tables S3–S6.

Safety Statement

No unexpected or unusually high safety hazards were encountered during this study.

RESULTS

Instructional Activity

As described earlier,⁷ the core of this activity centers on students using GenAI tools to draft essays on topics within their discipline. After composing their drafts, students engage in a systematic, instructor-supported evaluation of the AI-generated content, including references. To verify the authenticity and accuracy of each citation, students are recommended to use PubMed and Google Scholar. Toward the end of the course (ninth week), students present short in-class presentations where they share the challenges they encountered, effective strategies for verifying sources, and the ethical dilemmas they faced throughout the process. After these presentations,

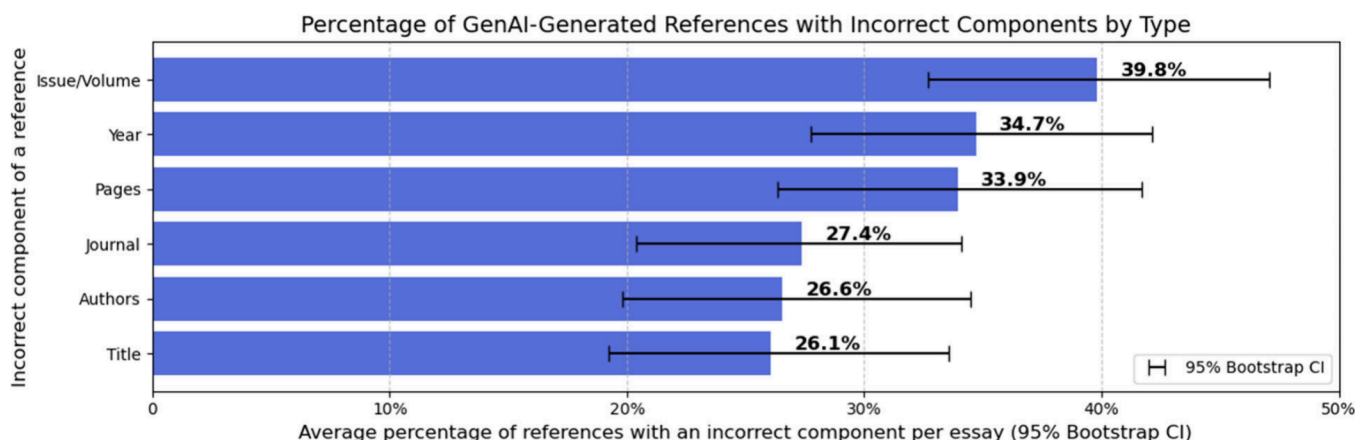


Figure 2. Frequency of inaccurate individual reference elements in GenAI-assisted student essays. References in each student essay were evaluated for correctness across six citation elements: author, year, title, journal, volume/issue, and page numbers. Bars represent the average proportion of incorrect reference components within an essay; error bars represent the 95% bootstrap confidence interval (CI). For example, the value of 26.1% for "Title" means that publication titles were cited incorrectly in 26.1% of references per essay, on average. The highest rates of inaccuracy occur in the "Volume/Issue" field, suggesting that these components are particularly susceptible to GenAI-related errors.

students are given an additional 3 weeks to complete the verification process and revise their GenAI-assisted essays to ensure accuracy and incorporate their own writing.

The learning goals for reference validation exercise include the following:

- Make research integrity visible and actionable by enabling students to directly confront and address the problem of fabricated references in their own work.
- Build students' information literacy and critical evaluation skills by requiring them to justify the reliability of each reference.
- Encourage dialogue about the responsible use of GenAI in academic contexts, including transparency about tool use, acknowledgment of AI limitations, and the ethical responsibility to independently verify sources before citing or sharing.

By guiding students in their use of literature sources and verification tools, this activity aims to enhance their information literacy, critical thinking, and understanding of research integrity, which are vital skills in the age of AI-augmented communication. This verification stage became the centerpiece for discussion and exploration of research integrity concepts. Students encountered a variety of articles, which did not exist, where author attributions were incorrect, or key details, such as publication year or journal name, were erroneous. Through guided analysis and group discussion, students developed strategies for identifying fabricated articles, learned the importance of cross-referencing multiple sources, and reflected on the ethical and practical consequences of incorporating unreliable citations in scientific communication. Over three semesters and 69 participating students, this activity provided a rich context to observe not just the significant number of fabricated citations in GenAI-assisted essays but also shifts in students' awareness of the importance of reference validation and correction.

Reference Accuracy: Real versus Fabricated

By analyzing each component of the references, we found that reference components that were incorrect in the majority of references were page numbers, issue/volume, and publication year (Figure 2). This trend aligns with the findings of Cheungpasitporn et al., who examined the accuracy of ChatGPT-3.5 in identifying relevant nephrology literature.³¹ In comparison to our study, Cheungpasitporn et al. did not explore inaccuracies associated with volume, issue, or page numbers and generally reported higher accuracy rates for the remaining reference components. Thus, incorrect journal names were found in 14% of references (compared to 27% in our study), incorrect years in 10% (vs 35% in our study), and incorrect authors in 3% (vs 27%). This discrepancy could be attributed to Cheungpasitporn's use of a single AI-tool and standardized prompt to generate all references, while our study involved students independently creating their own prompts using a variety of tools. We believe our approach more closely represents the real-world scenario, where individual users interact with GenAI tools in unique ways.

When we compared the actual publication years with those generated by GPT (Figure 3), we found the latter to display a systematic bias toward future dating. We hypothesize that this tendency in AI hallucinations may stem from training data sets that predominantly include open-access sources, which are often available online prior to their formal publication date.^{38,61,62}

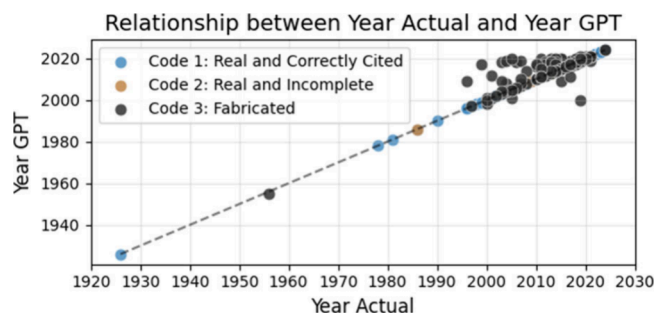


Figure 3. Relationship between actual and GPT-generated publication years. Each dot represents a single reference; its horizontal position reflects the true publication year and its vertical position the GPT-generated year. Dots are color-coded to indicate citation accuracy: blue for references that were real and correctly cited, brown for real references with incomplete citation information, and black for fabricated references. A fabricated reference is defined as having incorrect at least one of the six reference components (author, year, title, journal, volume/issue, or page numbers). Thus, 106 fabricated references still retain identifiable years and are included in this plot.

After analyzing the accuracy of individual reference components, all 456 AI-generated references were categorized into three groups. Of these, 53.7% ($N = 245$) were classified as Real and Correctly Cited (RCC), meaning all components were accurate; 1.5% ($N = 7$) were identified as Real but Incomplete (RI), where one or two of the six components were missing; and the remaining 44.7% ($N = 204$) were classified as Fabricated Citations (FC), indicating that at least one component was incorrect or GPT-fabricated. These results, counted per student essay, are summarized in Figure 4.

Note that definitions of fabricated and authentic references vary across different studies.^{31,34,35} In many cases, a reference is considered “existing” if it has a correct DOI, even if other components are incorrect. In our study, only 18.6% of references ($N = 85$) contained a DOI link, and of those, 64.7% ($N = 55$) were real. Acknowledging the subjectivity in distinguishing fabricated from nearly correct citations, we decided that any reference with even one incorrect component would be considered fabricated, as such citations should not appear in scholarly work. And although inconsistent definitions complicate direct comparisons, our findings are consistent with earlier reports. For example, Cheungpasitporn et al. analyzed 483 references generated by ChatGPT-3.5, reporting that 7% were incomplete and 20% were authentic.³¹ Our results show a similar rate of incomplete references (7%) and a much higher proportion of “authentic” or RCC references, averaging 51.3% per essay ($t(68) = 7.72, p < 0.001$, Cohen's $d = 0.93$).

In regard to fabricated references, we found a rate of 46.7%, compared to 31% reported by Cheungpasitporn et al., who used stricter criteria that required all six components to be incorrect.³¹ Applying their criteria in our earlier study,⁷ we found a fabrication rate of 26%, indicating higher success rate for our students. Other studies of AI-generated medical references reported fabricated references rates of 63.8% for ChatGPT-3,³⁴ 47% for ChatGPT-3.5,³⁵ and 56.7% for ChatGPT-4.³⁶

Students' Perceptions

It is also interesting to consider students' perception of reference fabrication rates.⁷ When self-reporting incorrect reference components, students estimated that titles, authors, and publication years had 39%, 35%, and 38% of errors, respectively. Yet, an objective data analysis of these same citations showed

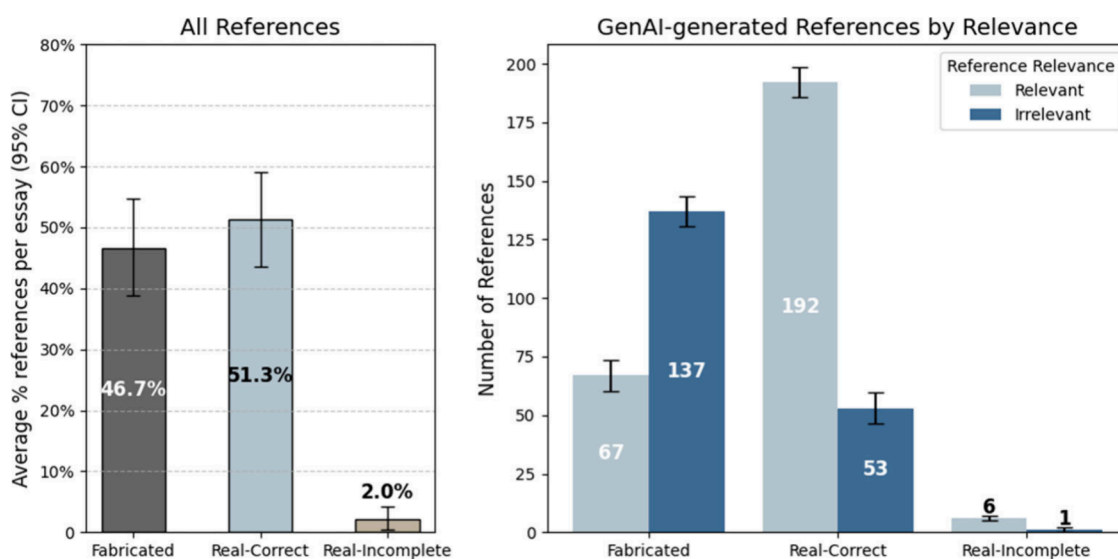


Figure 4. Quality of references (Real versus Fabricated and Relevant versus Irrelevant) in GenAI-assisted students' essays. (left) Average percentage of references per essay classified by accuracy: Fabricated (dark gray) vs Real and Correctly Cited (blue) vs Real but Incomplete (light gray). Error bars represent the 95% bootstrap confidence interval (CI). This graph shows that fabricated references account for approximately 46.7% of all citations in an average student's essay. (right) Categorization of references by both accuracy (real vs fabricated) and relevance (retained vs omitted in the final essay draft). Relevant references are depicted with light blue bars, while irrelevant ones are shown in dark blue. This graph shows that the majority of real and correct references were retained in final essays as relevant, whereas only about one-third of fabricated references were kept, either because students adopted them without verification or corrected them and found them useful to their topic.

somewhat lower incidence of errors. This mismatch underscores the limitations of relying solely on student self-reports for assessing citation accuracy.

Understandably, students' experiences with AI-generated citations often led to frustration and skepticism about the reliability of references. These two students described their challenges as follows:

One of the major issues that I noticed was with the references the AI used. The AI did not cite the sources correctly at all which made them nearly impossible for me to find. It had fully changed the names of some of the articles and authors in its references as well as inaccurately reported the DOI codes of the papers. Originally, I had thought that the AI had just made-up fake citations/articles since I could not locate any of the citations no matter what I tried. Eventually, after diving deeper into the topic, making corrections to the AI and working on my hot topic, I stumbled across one of the papers. Subsequently, I was able to find the other ones in the reference lists of each other and using key words combined with the first listed author and publication year.

A particularly frustrating issue was the accuracy of sources. When I requested citations, the AI provided references that were not only irrelevant but often fabricated. All of the papers it cited did not exist, and the titles were entirely incorrect. I was eventually unable to find any similar articles and eventually had to find new articles to support the essay. This made me realize that AI-generated content cannot be trusted to provide reliable or verifiable citations. As a result, I spent a substantial amount of time fact-checking nearly every statement in the AI-generated essays and searching for relevant articles, which undermined the efficiency of using the tool.

These accounts reflect a common sentiment among students, whose trust in AI-tools was undermined by inaccuracies and inconsistencies in citation information.

Relevant versus Irrelevant

When students were revising their GenAI-assisted essays, some AI-generated references were removed if deemed irrelevant, while relevant references remained and were corrected as needed. The relevance of a reference was determined individually by each student, and here is how one student describes their experience: "When evaluating these sources and where the AI cited them, some of them seemed to be a stretch. Some sources did not seem to match the information they were being used for. This prompted me to find better sources to represent the information generated by the AI."

Overall, among 456 total references, 58.1% ($N = 265$) were deemed relevant. As expected, a high proportion of relevant references were categorized as RCC and RI (74.7%). However, 32.8% of fabricated citations were also relevant. This is because 50.5% of FCs contain at least one real reference component that could enable students to deduce the original source. As in regard to students' self-perception of citations' relevance, students previously self-reported an average citation relevance of 63%.⁷ This figure slightly overestimates the total number of relevant citations but underestimates the relevance of RCC citations. In comparison, the citations' relevance reported by Cheungpasitporn et al. is 60.3%, though their decision criteria are unclear.³¹

AI Tool

Although students were required to use a chatbot for preparing their first draft of an essay, they were given an opportunity to choose any chatbot and any version of the tool they preferred. The recommended tools for the class were OpenAI's ChatGPT and U-M GPT. As a result, among 69 students,

- 30 students used U-M GPT-4.0;
- 25 used ChatGPT, an AI chatbot developed by OpenAI. Specifically, 9 students used the Generative Pretrained Transformer model GPT-3.5, 9 students used model GPT-4.0, and 1 student used ChatGPT o1-preview. The remaining students did not specify the ChatGPT model;

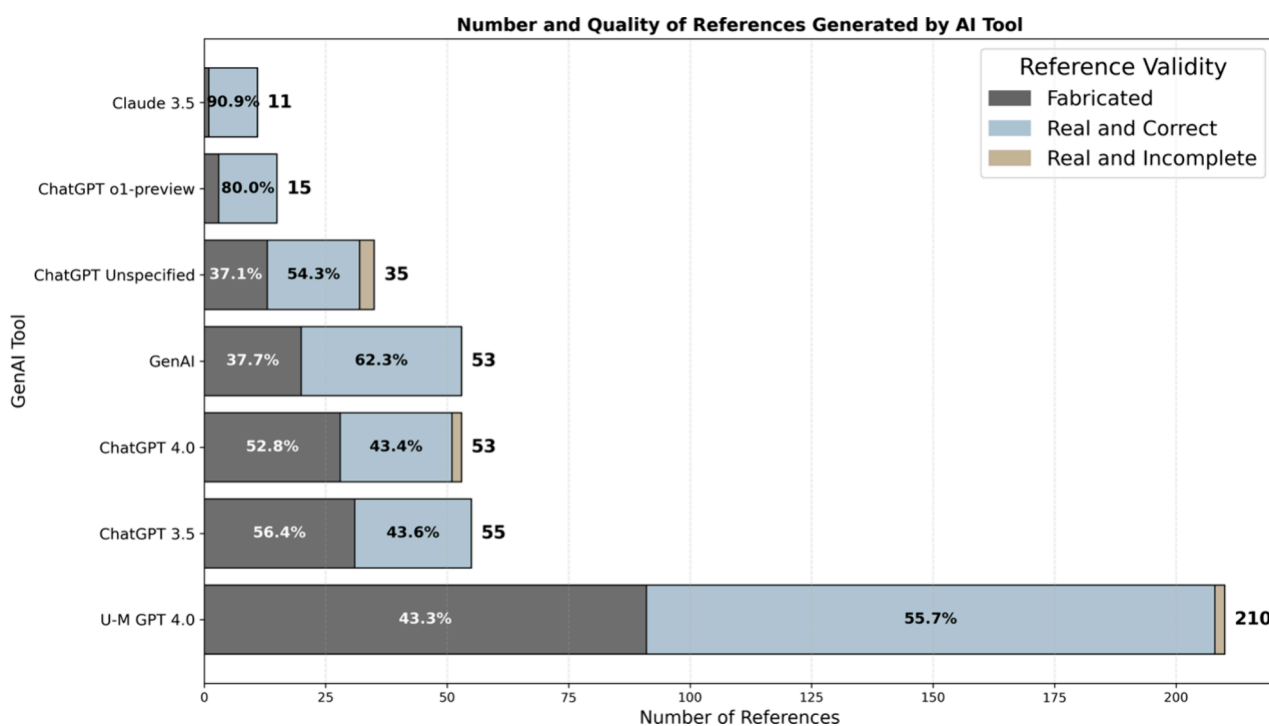


Figure 5. Distribution of AI-generated references based on the tools used to produce them. Here, references are grouped by the GenAI-tool and color-coded as dark gray (fabricated), blue (real and correct) or light gray (incomplete) to illustrate the proportion of correct and fabricated citations produced by each tool. Bar labeled as “GenAI” represents cases where students did not specify the tool utilized. The graph reveals that the majority of references were generated using the in-house tool, U-M GPT.

Quality of GenAI-generated References by Journal Type

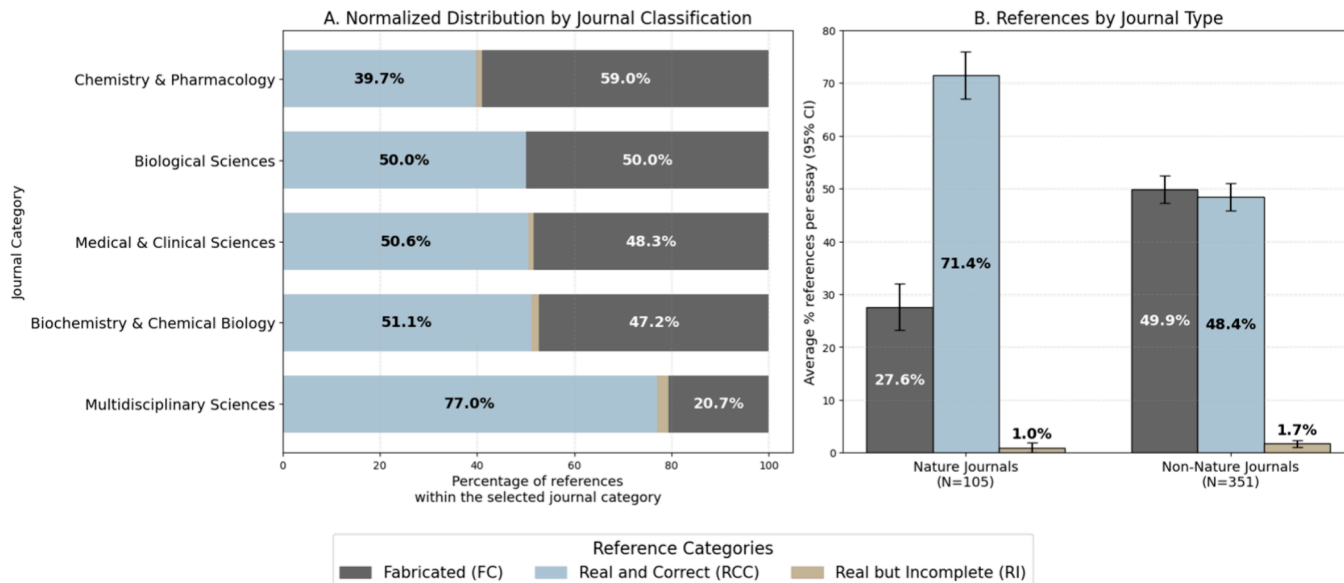


Figure 6. Quality of GenAI-generated references (Real versus Fabricated) by journal scope. Here, references are grouped by journal scope and color-coded as fabricated (dark gray), real and correct (blue), or incomplete (light gray), illustrating the proportion of correct and fabricated citations for each journal family. (A) Normalized distribution of fabricated and correct references within journal families classified by scope. The results reveal that multidisciplinary journals are associated with a higher proportion of correct references, whereas chemistry journals exhibit a majority of fabricated citations. (B) Comparison of references from *Nature*-family journals with those from non-*Nature* journals. This graph indicates that *Nature*-family journals contain a higher proportion of correct references and fewer fabricated citations compared to non-*Nature* journals.

- 1 student used Claude-3.5;
- 13 students did not specify the GenAI tool.

The most popular chatbot was U-M GPT 4.0; it was used by 30 students, who used it to generate a total of 210 citations, of

which 43.3% were fabricated. All of the tools produced fabricated citations, but the lowest fabrication rate was observed for Claude 3.5 (9.1%) (Figure 5). It is important to note,

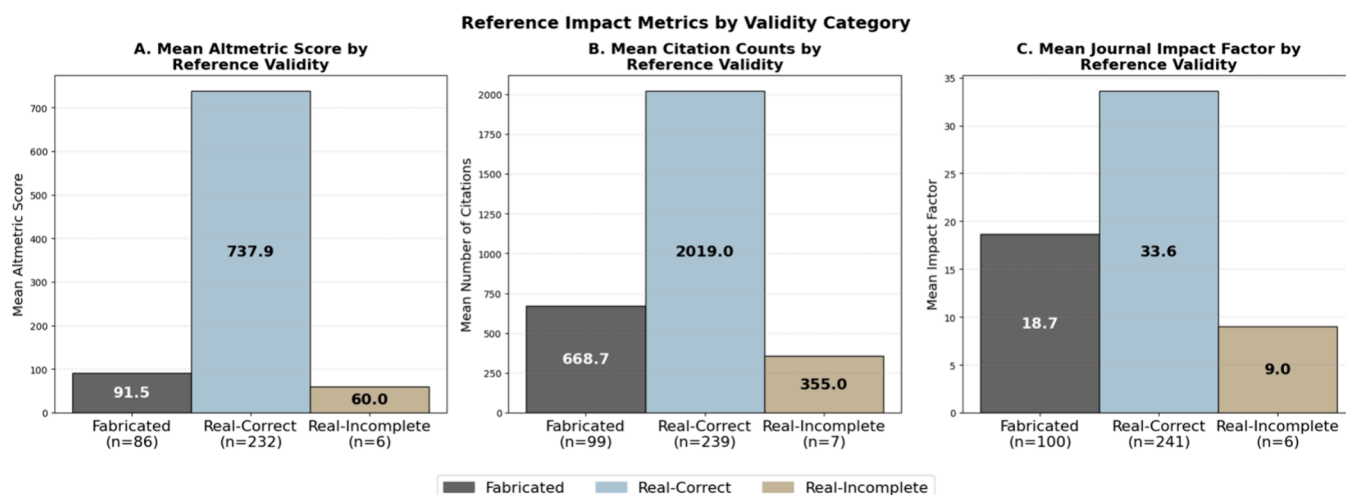


Figure 7. Journal and article parameters (mean) for three different types of AI-generated citations. Here, references are grouped by accuracy: fabricated (dark gray), real and correct (blue), or incomplete (light gray). (A) Mean Altmetric score for each reference group, highlighting social and mainstream media attention. (B) Mean citation count, reflecting scholarly impact across groups. (C) Mean journal impact factor for the cited sources. The results show that GenAI-generated references that are correct tend to be associated with the highest mean Altmetric scores, higher citation counts, and journals of greater impact relative to fabricated or incomplete references. See Table S2 for details.

however, that Claude 3.5 was used by only one student, so further testing is needed.

Journal Characteristics

When analyzing citations by journal type (see Table S1), we found that chemistry and pharmacology journals display higher rates of fabricated citations than any other type of journals. In contrast, multidisciplinary journals contributed the majority of correct references. Among the multidisciplinary journals, *Nature* stood out, outperforming all other journals (Figure 6), a result consistent with our previous study, which found a significant difference between *Nature* and non-*Nature* journals.⁷ Of the top 10 journals with the highest proportion of correct references, five were *Nature* titles (see p S5 in the Supporting Information (SI)).

To better understand what distinguishes particular journal types, we examined article citation metrics, including journal's impact factor and measures of article visibility, such as Altmetric Attention Score, Wikipedia mentions, and citation counts (Figure 7 and Table S2). Note that per our definition of a fabricated reference, it may include real components traceable to an existing article. Thus, 42% of fabricated citations (FCs) had Altmetric scores, as compared to 95% of RCCs. Similarly, journal impact factors are available only for 49% of FCs as compared to 98% of RCCs.

Altmetric scores showed considerable variation, ranging from 1 to 32,049, largely due to a few high-impact articles. Our analysis revealed that FCs had lower Altmetric scores as compared to RCCs, as confirmed by a Mann–Whitney test ($Z = -4.04$, $p < 0.001$). Additionally, *Nature*-type journals showed much higher median Altmetric scores than non-*Nature* journals (62 and 17, respectively), a difference that was statistically significant (Mann–Whitney test: $Z = -4.51$, $p < 0.001$). For Wikipedia mentions, we observed a similar trend, namely, RCC papers had on average a greater number of Wikipedia mentions compared to fabricated papers. However, a Mann–Whitney test indicated this difference was not statistically significant ($Z = 1.31$, $p = 0.189$). For citation counts, RCC papers were cited significantly more than fabricated papers (Mann–Whitney: $Z = -6.02$, $p < 0.001$). We also observed a significant difference between the median impact factors of journals containing RCC

articles and those linked to fabricated articles (Mann–Whitney: $Z = -4.18$, $p < 0.001$).

DISCUSSION

The integration of generative AI tools like ChatGPT into writing assignments presents both remarkable opportunities and distinct challenges for chemistry education. While these chatbots have the potential to enhance student learning,^{12,14,18,63} they also raise serious concerns regarding ethics, copyright and intellectual property, inherent biases, and the risk of generating and perpetuating misinformation.^{26,27,33,63} To address these complexities, a writing assignment that incorporated GenAI assistance was introduced in two chemistry courses: Bioinorganic Chemistry and Nucleic Acid Biochemistry. This assignment also served as an opportunity to make research integrity principles clear and actionable by emphasizing the importance of proper citation practices. Rather than viewing citations as merely a technical nuance of the writing process, students were asked to draft essays using GenAI-tools and then rigorously evaluate each AI-generated citation for factual accuracy and contextual relevance. Through this process, students learned that verifying references is not just a supplementary task; it is an important process that reinforces information literacy and research integrity, enabling the effective and responsible communication of knowledge.

The data we analyzed from student writing assignments across three semesters of chemistry courses illustrate the risks and responsibilities associated with referencing in the digital age. Out of 456 citations generated by AI on topics in bioinorganic and nucleic acid biochemistry, nearly half (44.7%) were fabricated, containing at least one incorrect component, most commonly erroneous issue, volume, and/or publication year (Figure 2). These results highlight concerns observed earlier in the analysis of GenAI-produced medical references,^{31,34,35} and reflect broader issues with reference accuracy.^{64–66} The introductory lecture by a chemistry librarian cautioned students about the likelihood of AI-generated references being fabricated. However, it was through students' own hands-on experience, experimenting with various prompts and investing considerable effort into validating references, that they became fully aware of

challenges inherent in GenAI-assisted referencing. Often, fabricated references with only a single incorrect element could be traced to a real source, while in some cases, just one accurate detail allowed students to deduce an existing reference. These “almost correct” citations served as a clear demonstration that research integrity involves more than simply avoiding outright fraud; it requires careful scrutiny of evidence, even when data appear partially correct or ambiguous. Verifying even seemingly plausible citations frequently required consulting multiple databases and cross-checking bibliographic information across platforms. On average, students reported spending about 1 h on the verification step, and more than 2 h combining self-written content with validated GenAI-generated materials (see Table S3). Overall, a great proportion of students commented negatively on their experiences in generating a valid bibliography (see Table S4). Many described initial frustration and disillusionment as citation after citation failed to correspond to a real article. As one student noted, after spending significant time checking DOIs, they realized that not a single AI-supplied reference for their topic actually existed. This sense of “betrayal” by AI led to a crucial shift: students moved from passive reliance on digital output to active, critical engagement with the literature. The cognitive discomfort of repeatedly encountering supposedly credible citations that could not be verified encouraged the development of more robust strategies for evaluating literature, including “source triangulation”⁶⁷ and paying closer attention to source relevance, all fundamental aspects of information literacy.

As students’ learning experience moved beyond simply verifying accuracy to evaluating the actual relevance of sources, they quickly realized that not all “real” references generated by AI were useful; in fact, only 58.1% ($N = 265$) were ultimately deemed pertinent to their essay topics. The remaining citations, although authentic, did not meaningfully support the specific chemical subjects under discussion. Note that students worked directly with references suggested by GenAI, thus bypassing a traditional literature search using scholarly databases. In this case, students had no judgment in selecting their initial set of references, for which they frequently use high number of article citations as a cue for the article’s relevance.⁶⁸ Instead, students had to critically assess each reference generated by AI and, if it existed, read its abstract, consider topical fit, and sometimes discard AI-supplied sources entirely. Peer discussions provided another opportunity for students to reflect on source selection more critically, which altogether fostered their information literacy skills, research integrity judgment, and disciplinary expertise, and produced coherent and trustworthy essays.

The tendency of generative AI to produce fake references stem from its propensity to hallucinate and its inability to accurately assess the validity or relevance of sources.^{63,69} Additionally, although little is known about the specific materials LLMs are trained on, we do know that much of their training data comes from scraped Internet text, which is inevitably biased, predominantly open-access, and often included without any consent.^{38,61,62} When exploring the underlying reasons for fabrication of chemical references in students’ essays, we noticed that journal type, impact factor, Altmetric score, and Wikipedia page mentions influenced the number of correctly cited articles. Specifically, multidisciplinary journals, high-impact journals, and articles with high Altmetric scores and frequent citation on Wikipedia had higher chance of being correctly cited, whereas articles with lower Altmetric scores, lower number of Wikipedia mentions, and published in lower impact factor journals or

disciplinary journals had higher chance of being fabricated. Similarly, Spennemann observed that the only authentic references generated by ChatGPT and DeepSeek were those also cited on Wikipedia pages.³⁸ These results also highlight an important discipline-specific finding for chemistry students. For example, references to high-profile multidisciplinary journals such as *Nature* were much more likely to be accurate. In contrast, disciplinary journals, which are arguably the most relevant for in-depth study of a disciplinary topic, showed higher rates of fabricated citations, reflecting GenAI’s uneven coverage and training data. This disparity has previously been observed in ChatGPT-generated citations on topics in environmental sciences, where ChatGPT overwhelmingly recommended publications in *Nature*-type journals.⁷⁰ The author, Eduard Petiška, attributed this trend to GPT’s excessive reliance on Google Scholar citation counts. By confirming a similar pattern in chemistry-related scholarly sources, our findings underscore the importance of fostering domain-specific AI literacy. Such training would enable students to distinguish between carefully curated disciplinary information from scientific databases and the broader, less selective content indexed by Google Scholar.

Despite ongoing improvements to ChatGPT and other advanced GenAI chatbots, they remain prone to hallucinations that stem from reliance on statistical patterns in imperfect and outdated training data. This can lead to underperformance due to mismatch between the outdated training materials and the current questions being asked.^{71,72} Recent research shows that retrieval augmented generation (RAG) can help mitigate some hallucinations,^{69,73,74} but it does not consistently improve reference accuracy or citation relevance^{75,76} and may even undermine the overall performance of LLMs.^{74,77} Given these persistent challenges, we believe that the instructional activity described herein will remain essential and relevant for fostering AI literacy and preparing students to uphold research integrity in the continually evolving landscape of AI-assisted scholarship.

Another notable finding from this study was the miscalibration between students’ perceptions of citation inaccuracies and the actual data. For example, students estimated the fabrication rate at 30%,⁷ whereas the true rate was 55%. Research on miscalibration in introductory biology courses has shown that this phenomenon is not uncommon.⁷⁸ Despite our efforts to set appropriate expectations about GenAI’s limitations through an introductory library lecture and a reference-checking exercise, students seemed to underestimate the number of fake references generated by GPT tools. This may be partially explained by a general tendency to remember positive outcomes.⁷⁹ However, a more troubling explanation involves cases of academic dishonesty that we, unfortunately, encountered in about half of student reports (49.3%, $N = 34$). Despite clear instructions to verify and correct the bibliography, some students included fabricated references in their final essays. These instances raise serious concerns about research integrity and highlight the challenges in maintaining scholarly standards in AI-augmented scientific communication, both in the classroom and beyond.⁸⁰

Overall, the reference validation activity demonstrates that, within chemistry education, research integrity can be effectively taught by directly involving students in encounters with GenAI-generated references. By directly observing how inaccurate citations disrupt the logical flow of ideas, weaken scholarly arguments, and undermine the integrity of scientific communication, students get confronted with a fundamental aspect of research integrity: the necessity of skepticism and independent verification. As a result, students learned that, regardless of the

convenience offered by emerging technologies, responsibility for citation accuracy and relevance ultimately rests with the author. While AI tools may increase efficiency, they cannot substitute for a scientist's duty to verify, contextualize, and ethically disclose every piece of evidence used. In this regard, despite the initial training, students felt that they wanted to learn even more about information validation, including how to verify information, read and understand articles, and critically assess sources (23%, $N = 5$; see p. S11 in the SI), as well as expressed interest in further exploring GenAI's limitations, such as ethical considerations, copyright, transparency, biases, and environmental impact (27%, $N = 6$; see p. S11 in the SI).

Several aspects proved critical to the success of this activity:

- An introductory lecture that described both the benefits and potential pitfalls of GenAI tools, helping students to set appropriate expectations.⁵⁶
- Additional instruction on effective verification strategies and the use of disciplinary databases to help students critically read and interpret scientific articles.
- Detailed prompt guidance provided through in-class materials and shared resources, which included analyses of sample prompts and discussions on why GenAI produced inaccurate or "fake" references.
- Peer-review feedback during in-class student presentations, where both peers and instructors challenged student-presenters to consider the relevance of their references within the broader context of their research topic and to reflect on their experiences with source validation.
- A requirement for students to submit drafts, which allowed for oversight of proper research conduct and opportunities for instructor feedback.²⁸

In summary, systematically exposing chemistry students to the limitations of GenAI fostered a more mature understanding of scientific integrity. Confronting fabricated and irrelevant citations required them to develop essential information literacy skills, sharpen their ability to judge relevance, and take ownership of the scientific record they help create. These lessons are fundamental not only for success in coursework but also for preparing students to uphold and advance trustworthy scholarship throughout their scientific careers.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available at <https://pubs.acs.org/doi/10.1021/acs.jchemed.5c01620>.

Journal and topic categorization parameters, student assessment, relevance scores, Tables S1–S6, and Figures S1–S4 (PDF, DOCX)

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Notes

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